

The Use of NN to Detect Learning Styles of Children with Learning Disabilities in E-Learning System

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Abstract. Every child learns differently and possesses a dominant or preferred learning style in certain ways. The cognitive development of children with learning disabilities (LDs) is often different than that of children without LDs. However, understanding how it differs from normal child development is important to understand how learning style identification can assist children with LDs. The aim of this paper is to propose an automatic learning style detection that works by the way of children interaction with the system. In this study, an Artificial Neural Network is used to predict the learning style of children with LDs.

Keywords: Learning disabilities · VARK model · Dyslexia · Learning style · Artificial neural network

1 Introduction

Learning styles are the most important and interesting element in the learning process since they may help learners become more self-aware of their strengths and weaknesses especially for children with learning disabilities (LDs) [1]. LDs refers to a disorder in one or more of the psychological processes involved in understanding or using spoken or written language. The term LDs does not involve learning problems that are primary the result of visual, hearing disabilities or mental retardation [2]. It is used to describe the most common types of learning disabilities such as dyslexia, dysphasia, dysgraphia, dyspraxia, and dyscalculia [3].

Children with LDs have specific cognitive abilities and different learning needs [4] and preferences [5]. The cognitive development of children with LDs is often different than that of children without LDs. However, understanding how it differs from normal child development is important to understand how learning style identification can assist children with LDs.

Various learning styles models and theories have been proposed to identify students' learning style and most of them focus heavily on the ability to think and how individuals acquire, retain, and retrieve information, and tend to collect style

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M. Ezziyyani (Ed.): AI2SD 2019, AISC 1102, pp. 210–216, 2020.
https://doi.org/10.1007/978-3-030-36653-7_21

dimensions to provide a profile of a learner's style [6, 7]. The traditional way to identify learning styles is to ask learners to complete questionnaires. However, these instruments present some disadvantages such as the lack of motivation to fill out a questionnaire and lack of self-awareness of their learning preferences or sometimes when learners fill out the questionnaire they feel bored because of the number of questions and they do not even understand the aim of the questions. To solve these limitations, numerous approaches for automatically detecting learning styles have been proposed. However, most of these approaches have been applied for adults or learners without LDs and did not include children with LDs. Hence, in this study we propose an automatic learning style detection for children with LDs.

This study tends to build a detection model of VARK learning styles. VARK learning style is one of the relevant methods used in e-learning systems and is characterized by the many learning objects or teaching materials available in online environment. We used an Artificial Neural Network to predict the learning style of children with LDs. We present the description of the proposed model and discuss some limitations, implications and research gaps that can be helpful to researchers working in the field of learning styles.

2 Related Work

Learning style can be defined as the composite of cognitive, affective, a psychological characteristic that serve as an indicator of how an individual interacts with and respond to the learning environment. Various learning style models and theories have been proposed to identify individuals learning style such as; *David Kolb's model*, *Peter Honey and Alan Mumford's model*, *Neil Fleming's VAK/VARK model*, *Anthony Gregorc's model*, *Felder–Silverman learning style*, and *Gardner's theory of multiple intelligences*. Every learning style model has its own assessment tool in a form of a questionnaire and it includes various questions about personality, study attitude and behavior.

Neil Fleming's VAK/VARK [7, 8] model is the most widely accepted and used model which defines how learners like to receive information. The model bases on the basic human observation channels; vision, hearing and feeling. VAK theory is considered to be one of the classical learning theories in the educational field, it defines three sensory modalities: Visual (see), Auditory (listen) and Kinesthetic (doing).

VARK is an acronym for Visual, Auditory, Read/write and Kinesthetic. It is a learning style model that has been modified from VAK model, to VARK learning style by Fleming in 2006. R is a second 'visual' modality for Read/write learners. According to [3], it seemed obvious that some students had a distinct preference for the written word whilst others preferred symbolic information as in maps, diagrams, and charts. These two preferences were not always found in the same person. There is more acceptance of this distinction today than in the 1980s.

Visual learning style (V) involves the use of seen or observed things. Auditory (A) learning style involves the transfer of information through sounds. Kinesthetic (K) learning styles involves physical and practical experiences.

Kolb learning style model [6] is one of the model includes *cycle of learning* as a central principle in his experiential learning theory expressed by the following stage of learning: abstract conceptualization (AC), concrete experience (CE), active experimentation (AE), and reflective observation (RO). Kolb's learning theory sets out four distinct learning styles types for which Kolb used the following terms: diverging (CE/RO), accommodating (CE/AE), converging (AC/AE), and assimilating (AC/RO).

Felder–Silverman learning style (*FSLSM*) [9] is the most popular model used in adaptive learning for adults, it become now a widely accepted learning style and is increasingly being adopted in many online learning systems. It describes four learning style dimensions, which indicate the student's preferences for certain poles of the dimensions; Active and reflective, sensing and intuitive, visual and verbal, sequential and global.

There exists a variety of research within the areas of LS detection and e-learning systems. Among the studies there is a study conducted by [10], the researchers presented an approach to recognize automatically the learning styles of individual students according to the actions that he or she has performed in an e-learning environment and the recognition technique was based on neural networks. Similar works can be found in the work of, the researchers proposed an approach using ANN to identify students' learning styles based on the Felder-Silverman learning style model (FSLSM).

3 Our Proposed Approach

The aim of this study is to offer to children with LDs a personal and a unique learning experience based on their learning preferences. For this purpose, we propose an adaptive approach for learning, which considers the learning preferences of children with LDs in e-learning systems. We used Artificial Neural Networks (ANN) to identify children with LDs learning styles based on the VARK learning style model. While there exist many learning style models, we selected the VARK model because it seems to be most appropriate for the use in educational systems for our target learners. According to [5] VAK learning style model uses the three main sensory receivers (vision, audition and kinesthetic) to determine a student's preferred or dominate learning style and provides a different perspective for understanding a student's strengths. Its use can be helpful with children with LDs to find out their learning styles and provide a general idea about their preferences in learning environments. Table 1 represents the description of the VARK learning style model and recommendations in e-learning system that support the learning style.

ANN are a type of machine learning that is inspired from the human brain. These days, ANN are able to perform what has been termed deep learning. An ANN is based on a collection of connected units or nodes called artificial neurons that transmit a signal from one neuron to another. The essential building block of ANN is a perceptron which accomplishes simple signal processing, and these are then connected into a large network.

Using ANN can help determine the learning style of a learner in both online and offline environments, unlike the traditional way of giving questionnaires to learner to be filled and infer the learning style from them. A Multilayer Perceptron (MLP) for multi-class Softmax Classification Neural Network has been proposed to classify the learning style of children. The network is composed of an input layer of two hidden layers and an output layer as shown in Fig. 1.

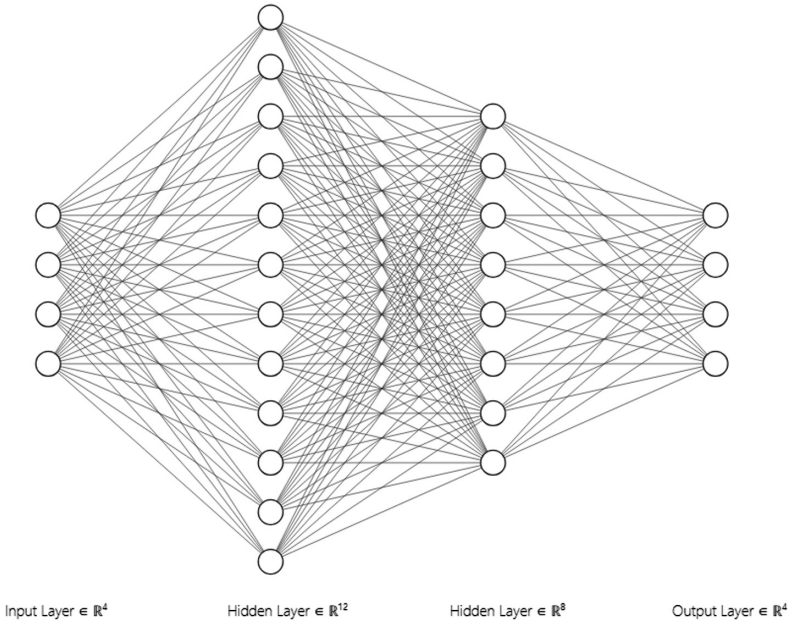


Fig. 1. The MLP model to classify the learning style of the learner.

Before feeding the data to the network a preprocessing step of normalizing the data, every feature linearly rescaled to the [0,1] interval using the min-max normalizer with the following formula:

$$z = \frac{x - \min(x)}{[\max(x) - \min(x)]} \tag{1}$$

Using Rectified Linear Unit (ReLU) as an activation function for each layer and the Softmax function for multi-classification. Relu function is calculated using the following formula $f(x) = \max(0, x)$ where x is the input to a neuron. Softmax function calculates the probability that any of the classes are true, it is calculated using the following formula:

$$\sigma(z)_j = \frac{e^{z_j}}{\sum_{k=1}^{k} e^{z_k}} \quad (2)$$

Where z is a vector of the inputs to the output layer and j indexes the output units. The model is initialized with random weights, in the forward propagation the network calculates the output and compare the result with the target, then the error is calculated.

$$Error = target - output \quad (3)$$

Since all the hidden units have contributed to the error in the output layer, the output error signals are transmitted backwards from the output layer to each unit in the hidden layer that immediately contributed to the output layer. This process is repeated layer by layer, until each unit in the network has received an error signal that describes its contribution to the overall error. Once the error signal for each node has been determined, the errors are then used by the unites to update the weights until the network converges to a state that allows all the training patterns to be encoded.

The output vectors for each type of Visual is [0 0 0 1], for Auditory it is [0 0 1 0], for Reading–Writing it is [0 1 0 0], and for Kinesthetic it is [1 0 0 0].

The proposed NN architecture aims to find a mapping between children with LDs actions in the e-learning system and the learning style that they best fit. To achieve this goal, we must identify the inputs of the network, its outputs and the meaning of their possible values as shown in Table 2. V1, V2, V3 and V4 are the inputs of the network, each one consists of the number of clicks or visits of an element such as videos, games, audio files, puzzles.

Table 1. Description of the VARK learning style model and recommendations in e-learning system.

Learning style	Online learning activities	Recommendations in e-learning system that support the learning style
Visual-V Preference for graphical and symbolic ways of representing information	Visual materials like video, visuals of sample cases and scenarios, simulations, graphic demonstration and presentation	3D tools, drawing Online, interactive News, virtual tours Video conferencing
Auditory-A Preference for oral information	Audio materials like podcasts of sample cases and scenarios, narrated tutorials verbal presentations	e-books, voice over Internet, guided Reading
Reading/writing RW preferences for information printed as words	Reading and writing Materials like readable documents	Chat Messaging course Notes
Kinesthetic-K Preference related to the use of experience and practice	Kinesthetic and tactile materials like virtual reality systems	Flash animations, Online Simulation tools Tangible learning systems and Virtual labs

Table 2. The inputs and outputs of the NN.

Inputs	Description	Outputs
V1	N° PPT presentation, slides N° diagrams N° graphs—posters N° charts	V
V2	N° audio e-book N° verbal presentation N° song and sound effects N° reading and spelling programs	A
V3	N° notes—text N° glossaries N° dictionaries N° handouts	R
V4	N° games N° virtual simulation N° interactive presentation N° flash animation	K

4 Conclusion and Future Works

In this paper, we proposed an adaptive approach for learning, which considers the learning preferences of children with LDs in e-learning systems. We used ANN to identify children with LDs learning styles based on the VARK learning style model. For future work, we are planning to collect the data from questionnaires to train and test the network, the questionnaires will be answered by the children, and from the result of the questionnaires will assign one of the four classes based on the number of clicks or visits to the elements of each input.

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